

Math 428, Spring 2024, Homework 2 Solutions

Section 4.2: 2

Solution. $\frac{1}{(z+2)(z-3)}$ is analytic on $E = \mathbb{C} \setminus \{-2, 3\}$. Γ satisfies $\text{ind}_{\Gamma}(z) = 0$ for all $z \notin E$. So $\int_{\Gamma} \frac{1}{(z+2)(z-3)} dz = 0$.

Section 4.2: 3

Solution. Part. frac's: $f(z) = \frac{1}{(z+2)(z-3)} = \frac{1}{5} \left(\frac{1}{z-3} - \frac{1}{z+2} \right)$. We use the fact that $\int_{\gamma} \frac{1}{z - z_0} dz = 2\pi i \text{ind}_{\gamma}(z_0)$ to find

$$\int_{\gamma_1} f(z) dz = 0, \quad \int_{\gamma_2} f(z) dz = -\frac{2\pi i}{5}, \quad \int_{\gamma_3} f(z) dz = \frac{2\pi i}{5}$$

Section 4.2: 6

Solution. We use the Cauchy Integral Formula, and the fact that

$$\text{ind}_{\gamma_1}(\tfrac{1}{2}) = 1, \text{ind}_{\gamma_1}(\tfrac{3}{2}) = \text{ind}_{\gamma_1}(3) = 0, \quad \text{ind}_{\gamma_2}(\tfrac{1}{2}) = \text{ind}_{\gamma_2}(\tfrac{3}{2}) = 1, \text{ind}_{\gamma_2}(3) = 0$$

to find

$$\int_{\Gamma} \frac{e^z}{z - \frac{1}{2}} dz = 0, \quad \int_{\Gamma} \frac{e^z}{z - \frac{3}{2}} dz = 2\pi i e^{\frac{3}{2}}, \quad \int_{\Gamma} \frac{e^z}{z - 3} dz = 0.$$

Section 4.3: 4

Solution. For $z \neq 0$,

$$z^{-3}e^z = \frac{1}{z^3} \sum_{j=0}^{\infty} \frac{1}{j!} z^j = \sum_{j=0}^{\infty} \frac{1}{j!} z^{j-3} = \sum_{k=-3}^{\infty} \frac{1}{(k+3)!} z^k$$

Section 4.3: 5

Solution. We use the expansion for $|z| < 1$: $\log(1+z) = \sum_{j=1}^{\infty} \frac{(-1)^{j-1}}{j} z^j$, which comes

from $\log(1+z)' = (1+z)^{-1} = \sum_{j=0}^{\infty} (-1)^j z^j$, and $\log(1) = 0$. Then

$$\frac{\log(1+z)}{z^4} = \sum_{j=1}^{\infty} \frac{(-1)^{j-1}}{j} z^{j-4} = \sum_{k=-3}^{\infty} \frac{(-1)^{k-1}}{k+4} z^k$$

Section 4.3: 6

Solution. One way (one can also start with partial fractions):

$$\begin{aligned}\frac{z}{z^2+1} &= \frac{1}{z-i} \frac{z}{z+i} = \frac{1}{z-i} \left(1 - \frac{i}{z+i}\right) = \frac{1}{z-i} \left(1 - \frac{i}{2i+(z-i)}\right) \\ &= \frac{1}{z-i} \left(1 - \frac{\frac{1}{2}}{1 - \frac{i}{2}(z-i)}\right) = \frac{1}{z-i} \left(1 - \frac{1}{2} \sum_{j=0}^{\infty} \left(\frac{i}{2}\right)^j (z-i)^j\right) = \frac{\frac{1}{2}}{z-i} - \sum_{k=0}^{\infty} \frac{i^{k+1}(z-i)^k}{2^{k+2}}\end{aligned}$$

A1 Solution. The series converges uniformly on the image of γ , so we can interchange summation and integration to write

$$\int_{\gamma} f(z) dz = \sum_{k=-\infty}^{\infty} a_k \int_{\gamma} z^k dz$$

If $k \geq 0$, then z^k is analytic (on all of $\mathbb{C}$), so $\int_{\gamma} z^k dz = 0$ by Cauchy's theorem.

If $k \leq -2$, then we can write $z^k = \left(\frac{z^{k+1}}{k+1}\right)'$, so by the fundamental theorem of calculus and the fact that γ is closed we have $\int_{\gamma} z^k dz = 0$.

Thus, the only term that is not zero is $k = -1$. So we get

$$\int_{\gamma} f(z) dz = a_{-1} \int_{\gamma} \frac{1}{z} dz = a_{-1} 2\pi i \operatorname{ind}_{\gamma}(0).$$